Kerala Home, Phase 1

Pre-construction drawing set, Guntur, Andhra Pradesh (agricultural land: conversion to be confirmed)

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Project data

Site	Guntur, Andhra Pradesh; agricultural land (conversion to confirm)
Plot	27' x 75' = 2,025 sq ft (planned on this plot alone)
Built footprint	62' x 22' = 1,364 sq ft (wall centrelines)
Carpet area	1,267 sq ft clear, incl. poomukham and open wet area
Construction	RCC frame, AAC walls, steel-truss roof with clay tiles
Plinth	FFL +2'-0" above finished ground

Key design decisions

• Site is in Guntur, Andhra Pradesh (farm land), not Kerala; Kerala style kept, construction adapted to local climate and soil.

• Plot 27 ft x 75 ft (5 ft longer than first assumed; confirm by survey), road on east; house 62 ft x 22 ft (lengthened 2 ft for Bedroom 2) with an 8 ft front yard and 5 ft rear garden; owner accepts local practice on side setbacks (to be checked).

• RCC frame with AAC block walls: 9 in outer, 5 in inner, 7 in where plumbing is concealed.

• Plinth 2 ft above finished ground; poomukham 6 in lower; bathrooms 12 mm lower with no raised threshold.

• AC provisions (points, sleeves, drain, outdoor-unit brackets) in both bedrooms, the office and living; passive cooling first.

• Flat insulated false ceilings in bedrooms, offices, baths and kitchen; sloped ceilings in living, dining and verandah.

• No glass skylight: the dining gets borrowed daylight from the living, the nook window and the hall; a louvred ventilator lantern above it lets hot air out (no glass, so no greenhouse heat).

• Courtyard removed; the dining hall at plot centre keeps the Brahmasthan open, with the ventilator above.

• Single storey, no internal stair; roof access by fixed ladder; water tank on stand in SW rear garden.

• Service band on south (heat buffer); bedrooms and living get north light.

• Kitchen in SE; stove on east wall; open wet area onto the walkway; provisions in bench drawers, kitchen cabinets and lofts (no separate store).

• Bedroom doors widened to 3 ft; bathroom, office and store doors 2 ft 6 in.

• Twenty-fifth review: main door moved 3'-4" south (y 17'-7" to 20'-7"), centred under the front gable and opening straight onto the 5 ft walk between the pooja and the diwan; shoe cupboard to its north on the east wall; window diwan back to 6 ft along the north window, out of the door's line; steps, middle verandah pillars and the swing realigned.

• Twenty-fourth review: planned on the 27 ft plot alone (north strip dropped for now): roof eaves cut from 2'-6" to 2'-0" so the north eave stays inside the plot line, septic tank moved under the two-wheeler bay in the front yard with a lined soak pit beside it, all drains rerouted along the south walkway to the front, rainwater to the sump with overflow to the road; north-strip trees and recharge pit removed.

• Twenty-third review: the fridge moves out of the kitchen into the dining alcove beside the kitchen door, against the wet-area wall; the kitchen's north wall is now free.

• Twenty-second review: the kitchen's tall pantry and the dining table and chairs are removed at the owner's request; the dining space under the ventilator stays open (the pendant light and points stay for a table later).

• Twenty-first review: the open hall becomes three zones: family TV lounge in the dining bay (5'-6" sofa facing the TV), dining under the ventilator moved 9 in south (about 2 ft behind the chairs on both sides), and guest sitting by the main door with a 4'-6" window diwan (shortened after the walk-through showed a 1'-6" pinch at the shoe cupboard) and an armchair under the living's north window; a clear walk of about 4'-11" from the main door to the dining and kitchen, pooja front kept clear.

• Twentieth review: Bedroom 2's bed moves to the east wall, head east, leaving an 8 x 14 ft open floor from the door to the bath and window; wardrobe to the west wall (sound buffer to the master), work table north of the bed, AC on the bath's outer wall; dresser dropped (mirror on the wardrobe).

• Nineteenth review: with Bedroom 2 at 17 ft, its bed turns head-south in the middle of the room (2 ft+ on both sides, under the jali); Bath 2 grows to 7' x 7' (6'-7" x 6'-4" clear) with a 3' x 3'-6" wet zone, basin in the dry zone and the door on its east wall; a dresser takes the west wall; work table and wardrobe stay on the east wall.

• Eighteenth review: house lengthened 2 ft to 62 ft, all of it added to Bedroom 2 (now 17 ft wide; the sitting area below it also gains 2 ft); front yard 10 ft to 8 ft (check the front setback). Bedroom 2 window back to 6 ft.

• Seventeenth review: sitting area gets a 5 ft tall storage cupboard on the south wall between the bench and the window; the hand-wash basin moves to the east end of the serving counter, just outside the common bath, for washing up after meals.

• Sixteenth review (independent Indian-architect review): Bath 2 re-planned 7' x 6' so Bedroom 2's bed (head west) has 1'-6" on both sides, work table to the east wall. Kitchen entered only from the dining-side door (hinged, 3 ft); the north sliding door is walled up so the pooja front is quiet, and a tall pantry takes the north wall. East kitchen window cut to 2 ft so the hob and chimney sit on solid wall. Septic tank moved 6 ft+ from the house and the soak pit replaced by a soak trench. Ensuite door opens outward; common bath WC side clearance 17 in, door rehung at the north jamb with a closer, geyser and shower off the window. DB moved to the dining face of Bedroom 2's east wall.

Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Cover and sheet index**
Sheet	**A-000**
Scale	-
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Site conditions (Guntur)

• **Heat:** summer days 40-45 C (March-June). Insulated roof, ventilated attic, shaded west and south walls, AC provisions.

• **Rain:** about 850-1000 mm a year, mostly June-October, with short cyclonic downpours in Oct-Nov. Deep eaves, gutters, drip grooves.

• **Wind:** coastal cyclone belt. Trusses bolted down; every edge tile clipped; no light sheeting that can lift.

• **Soil:** black cotton (expansive) soil is common. A soil test decides the foundation; keep water and trees away from the walls.

• **Earthquake:** seismic zone III. RCC frame with lintel and ring bands.

• **Water:** groundwater is often hard with high fluoride. RO for drinking, softener if needed, matt fittings.

• **Termites and mosquitoes:** anti-termite treatment before the plinth; mesh on every opening.

• **Land:** agricultural land. Confirm conversion and the permission route before work starts.

Levels and heights

Level	Above finished ground
Finished ground (FGL)	+0'-0"
Poomukham floor	+1'-6"
Wet area floor	+1'-9"
House floor (FFL)	+2'-0"
Bathroom floors	+1'-11.5" (12 mm drop)
Door head / lintel band	+9'-0"
False ceilings (rooms)	+11'-6"
False ceilings (baths)	+10'-6"
Wall top / ring beam	+12'-0"
Roof eaves (tile edge)	+10'-10"
Main ridge	+18'-4"
Front gable ridge	+15'-5"

Wall types

Type	Build-up	Where
EXT	200 mm AAC + 20 mm external plaster + 12 mm internal plaster (~9 in)	Outer walls, kitchen north wall
WET	150 mm AAC + plaster both faces (~7 in): concealed plumbing walls	Walls with concealed plumbing
INT	100 mm AAC + plaster both faces (~5 in)	Other inner walls

General notes

• Dimensions are to wall centrelines unless marked 'clear'. Clear room sizes are on A-104.

• Concrete, steel, foundations and member sizes are by the structural engineer. Column positions are not shown and must be coordinated with the furniture layout.

• AAC: 150 mm galvanised mesh over every block-concrete joint; chases cut by machine; no hammering.

• All external sills, sunshades and copings get drip grooves; every external opening gets mosquito mesh.

• Wet rooms: waterproofing on floors, 12 in up the walls and 7 ft in showers; 48 h flood test before tiling.

• Skirting is flush with the plaster (no dust ledge). Grout in wet areas is epoxy.

• Pipes and conduits are laid before plaster and photographed with a tape measure in view, for future drilling.

• Keep 3 ft paved apron around the house, sloped away. No gardens, soak pits or big trees within 6-8 ft of walls.

Hold points (stop and inspect)

Stage	Check before it is covered up	Who
Before design freeze	Soil test report, level survey of plot and road, water quality test (TDS, hardness, fluoride)	Owner, engineer
Excavation	Footing depth and bearing layer seen in the pit; no black soil left under footings	Structural engineer
Before each concrete pour	Reinforcement, cover blocks, laps and column positions checked; photos taken	Engineer / site supervisor
Plinth	Anti-termite treatment (certificate), fill compaction, DPC laid and cured	Supervisor
Before plaster	Conduits, pipes, AC sleeves, mesh over AAC joints; photos with a tape measure in view	Owner, supervisor
Wet rooms	Waterproofing coats, upturns, 48 h flood test passed before any tile is laid	Owner, plumber
Roof	Truss holding-down bolts, membrane and insulation, edge tiles clipped, flashings at the ventilator and valleys	Engineer, owner
Before handover	Water test on every floor slope and sill, plumbing pressure test, electrical insulation and earth test	Owner, MEP

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Guntur, Andhra Pradesh

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Drawing	**Design basis and general notes**
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Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east



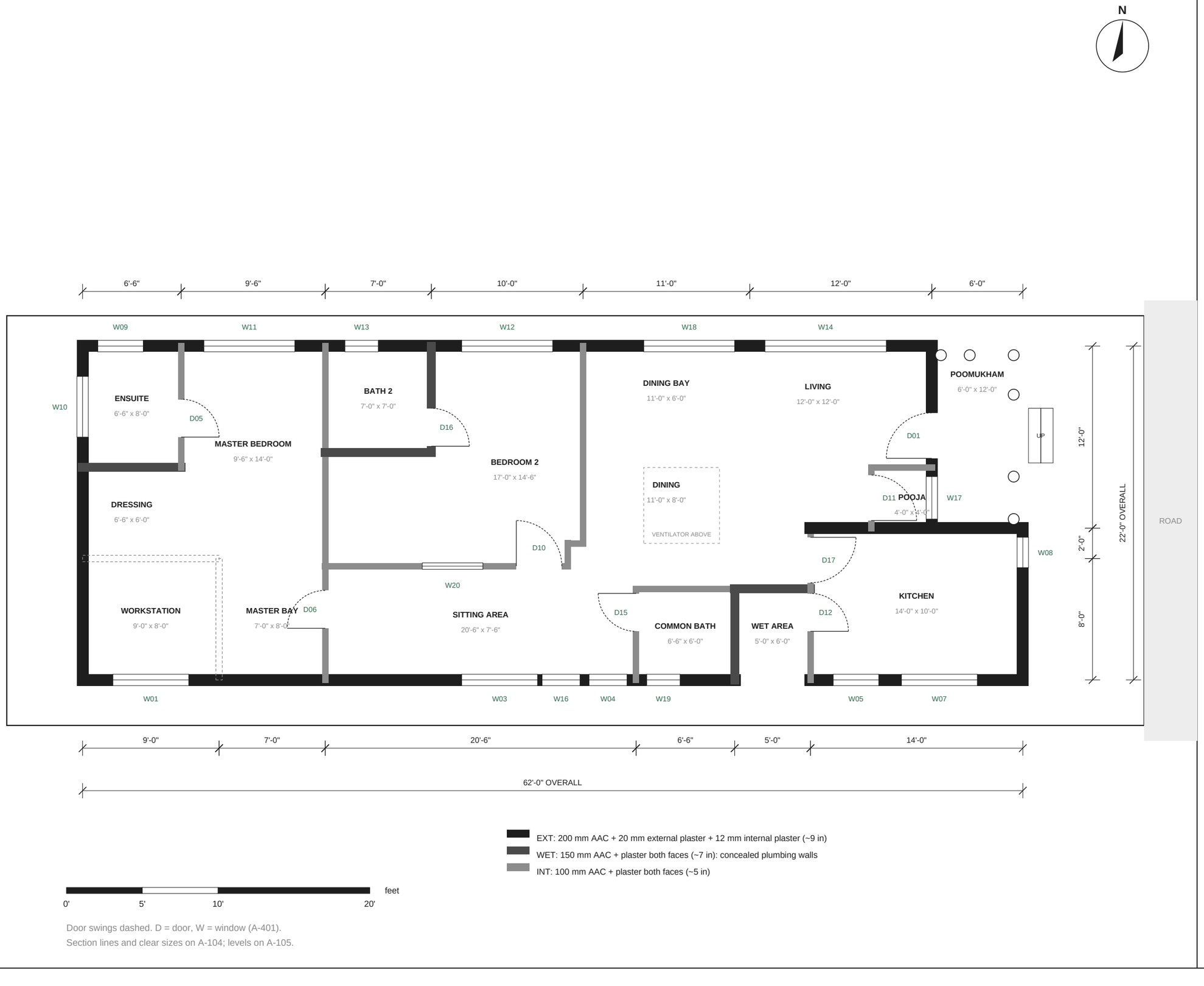
- TANK: Overhead tank on 18 ft stand (SW), ladder to roof
- SEPTIC: Septic tank, 2 chambers, sealed RCC, under the two-wheeler bay (front yard, SE), heavy-duty covers
- SOAK: Lined soak pit for septic effluent and grey water, front yard SE, as far from the house and sump as the plot allows (or treated outflow to a road drain)
- SUMP: Sump (NE)
- BORE: Borewell (if any), NE
- PERGOLA: Creeper pergola for west shade (no big trees near walls)
- TULSI: Tulsi thara
- BAY: Two-wheeler bay
- GATE: Gate
- STEPS: Entrance steps (3 risers), between the pillars, SS handrail one side
- FEET: Foot-wash tap + drain
- WSTEP: Steps from the open wet area down to the walkway (4 risers of about 5 in, along the wall)
- APRON: Plinth protection apron 3 ft, sloped away
- Site drains: finished ground falls 1:100 away from the house to the road side.

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Stage	Draft for professional review
Drawing	**Site plan and services**
Sheet	**A-101**
Scale	1:100 at A3
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Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east



Door swings dashed. D = door, W = window (A-401).
Section lines and clear sizes on A-104; levels on A-105.

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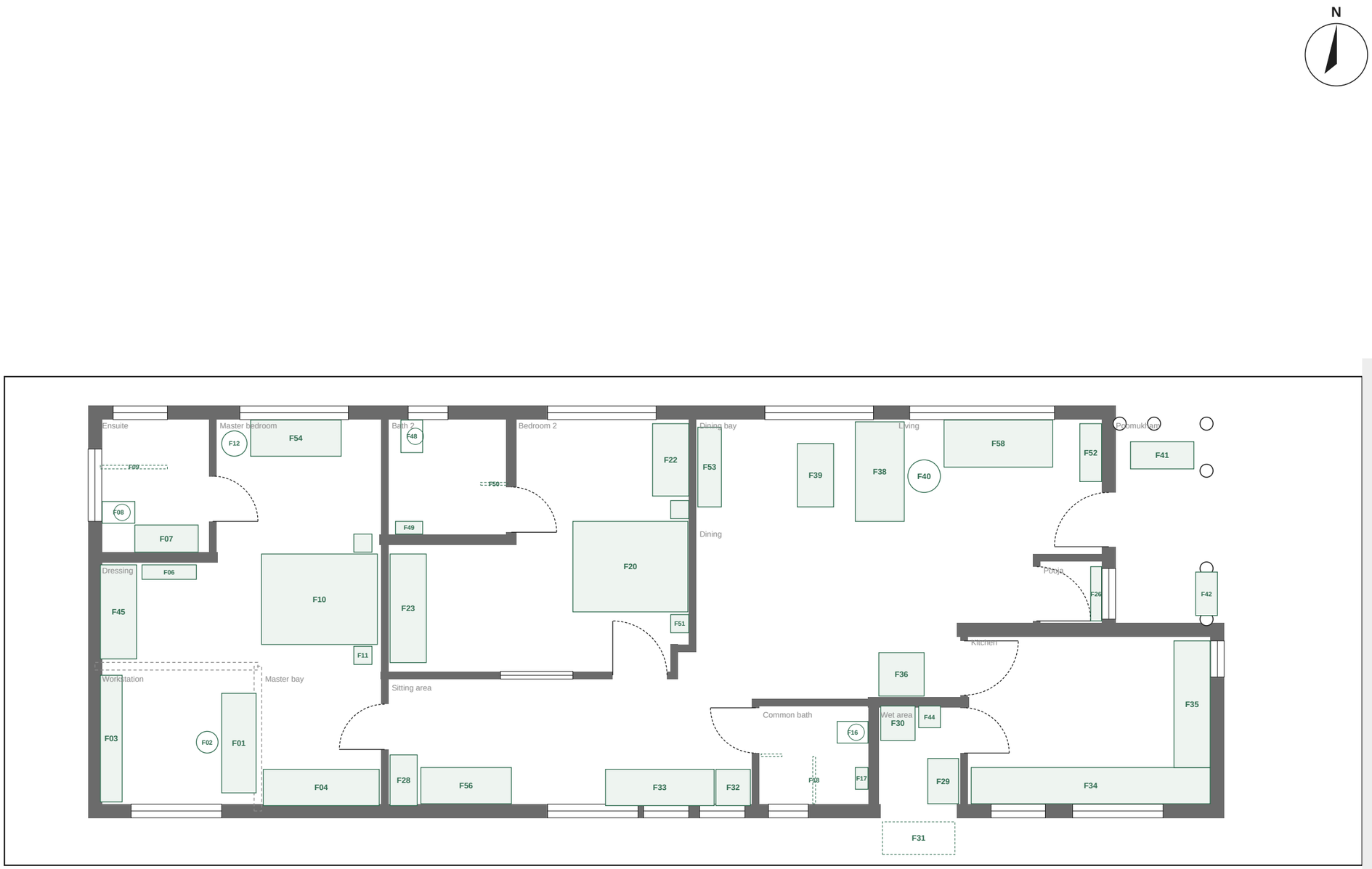
Stage	Draft for professional review
Drawing	**Ground floor plan**
Sheet	**A-102**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
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Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east



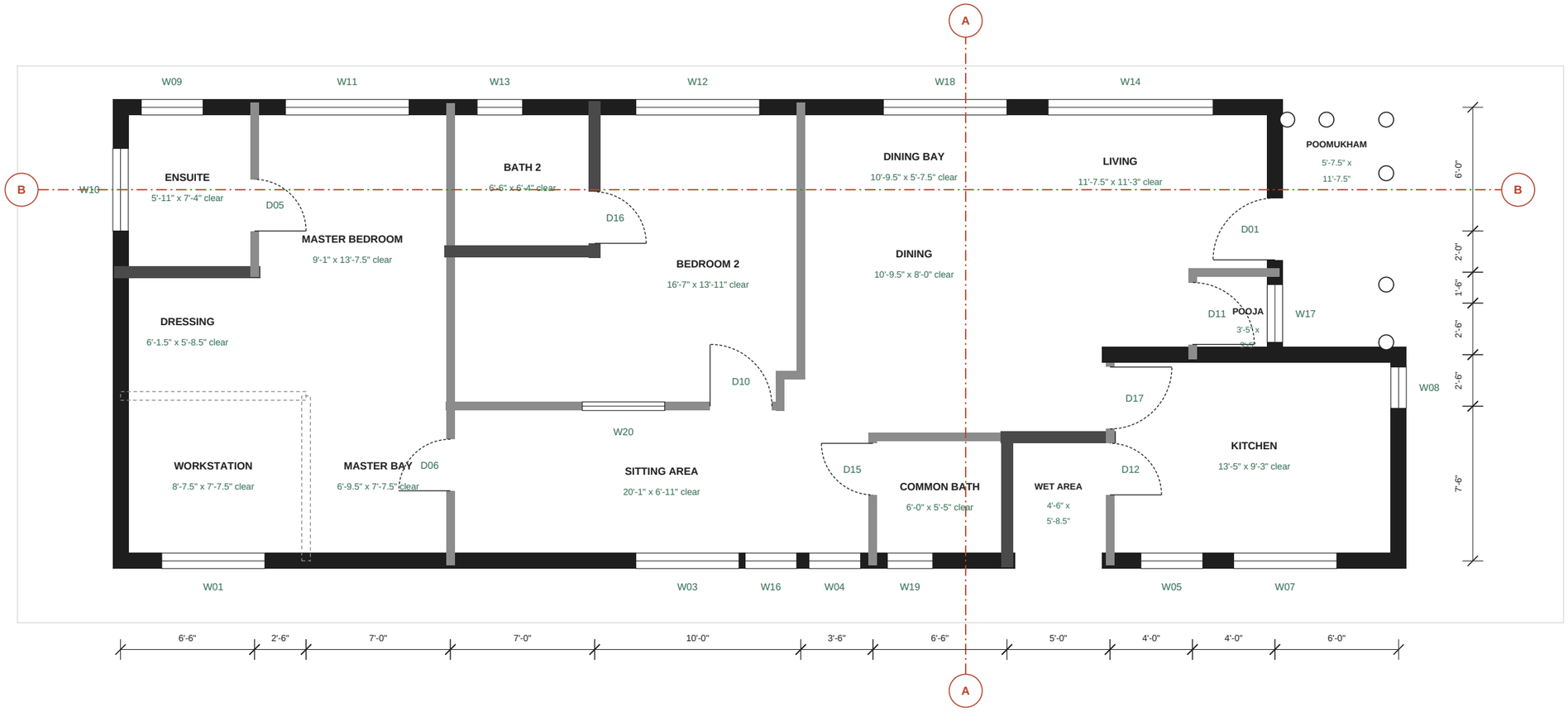
0' 5' 10' 20' feet

Furniture key (full specs in schedules)

- **F01** Work desk 5'6" x 2' (free-standing, cable tray)
- **F02** Ergonomic task chair
- **F03** Bookshelf, full height (video backdrop)
- **F04** Built-in wardrobe 6'-5", south wall, sliding doors, lofts above
- **F06** Dresser with mirror (north wall)
- **F45** Built-in wardrobe, west wall, sliding doors
- **F07** Vanity basin counter 3'-6"
- **F08** WC (EWC)
- **F09** Shower glass partition
- **F10** Queen bed 5' x 6'6" + headboard
- **F11** Bedside table (x2)
- **F12** Reading chair
- **F54** Window seat / diwan 5' x 2' under the north window, drawers below
- **F16** WC (EWC), 17 in clear to the north wall
- **F17** Wash basin
- **F18** Shower glass partition (3 x 3 ft shower in the south-west corner)
- **F48** WC (EWC), 15 in clear each side of its centre
- **F49** Wash basin (south wall, dry zone)
- **F50** Fixed glass splash panel at the south edge of the 3' x 3'-6" wet zone (north-east)
- **F20** Queen bed 5' x 6'6" + headboard, head to the east wall (padded headboard: TV on the other side)
- **F51** Bedside table (x2)
- **F22** Work table 4' x 2' with monitor (east wall, north of the bed; north window to the side)
- **F23** Built-in wardrobe 6' with a mirror door (west wall, shared with the master: also a sound buffer)
- **F26** Pooja mandapam on the east wall (idols face west; family faces east)
- **F28** Built-in bench with storage drawers below
- **F29** Utility sink counter (on the kitchen wall)
- **F30** Washing machine bay 2' x 2' (covered back corner, 2 in clear of the bath wall)
- **F31** Fold-away clothesline in the south walkway
- **F44** Tall slim utility cupboard (brooms, mops, detergents)
- **F32** Hand-wash basin at the east end of the serving counter, just outside the common bath (after meals); mirror on the side wall
- **F33** Serving counter with crockery cabinets above, 6 ft
- **F56** Tall storage cupboard 5' x 2' (bedding, suitcases, festival vessels), loft above
- **F34** L-counter, south run
- **F35** L-counter, east run (hob centred 7'-6" from the south wall face, on solid wall)
- **F36** Refrigerator space (dining alcove beside the kitchen door, against the wet-area wall, door facing north)
- **F38** Sofa 5'-6" (2-3 seats) facing west to the TV
- **F39** Coffee table 3'6" x 2'
- **F53** TV unit 4'-5" on Bedroom 2's east wall (55-65 in TV wall-mounted above)
- **F40** Armchair at the west end of the window diwan
- **F58** Window diwan 6' x 2'-6" under the north window, drawers below (guest seating, day bed); out of the line of the main door
- **F52** Shoe cupboard 3'-3" on the east wall just north of the main door (closed shutters, louvred for air)
- **F41** Oonjal (swing) 3'6", north side of the verandah, clear of the entrance
- **F42** Arugu seat (built-in, south end)

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Stage	Draft for professional review
Drawing	**Furniture layout**
Sheet	**A-103**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground



Clear sizes are wall face to wall face after plaster (green). Grid dimensions are to wall centrelines. Section A-A and B-B on A-301. Check every furniture item against the clear size before ordering. Column positions (by engineer) must avoid door swings, wardrobes and the TV wall.

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

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Stage	Draft for professional review
Drawing	**Working plan: clear sizes and sections**
Sheet	**A-104**
Scale	1:75 at A3
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Guntur, Andhra Pradesh

Plot 27' x 75', road on east



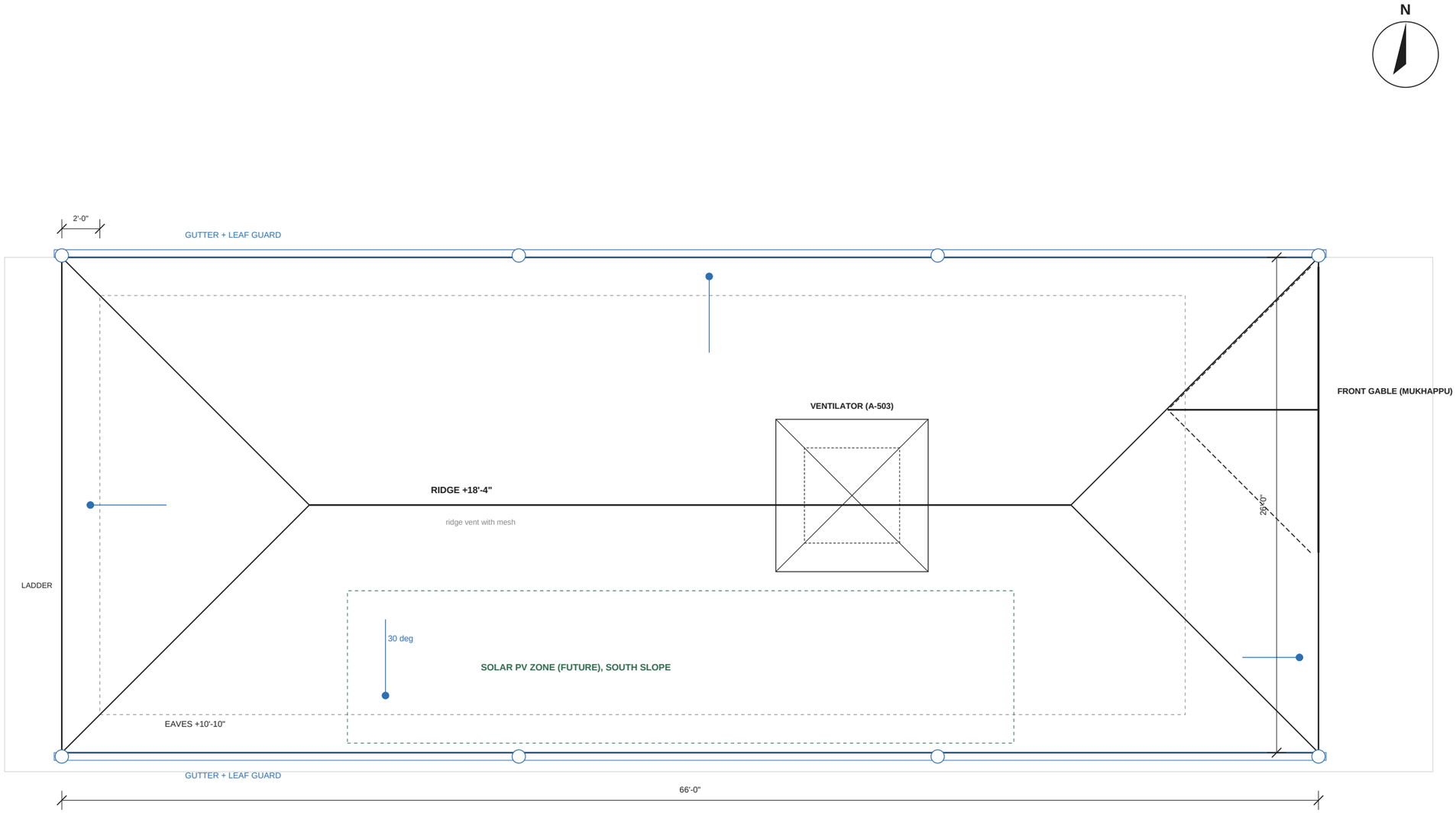
- Red oxide / Kota stone
- Matt vitrified 2x2
- Marble / teak (pooja)
- Anti-skid matt vitrified
- Anti-skid ceramic, waterproofed
- Rough Cuddapah / Kota, falls
- Flamed granite

Floor detailing

- Skirting 4 in high, set flush with the plaster face (plaster stops in a groove above it), so no ledge collects dust.
- Bathrooms 12 mm below the bedroom floor with a sloped granite threshold: no raised step to trip on, water stays in.
- Shower zone: linear drain at the far end, floor falls 1:60 only inside the wet zone; dry zone floor nearly flat.
- Poomukham and steps: flamed or leather-finish stone. Polished granite is slippery when wet.
- Kitchen: mid-tone matt floor; white glossy tile shows every stain and is slippery with oil.
- Leave 5 mm expansion gaps at walls under the skirting and every 15-20 ft in large floors.

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Stage	Draft for professional review
Drawing	**Floor finishes and levels**
Sheet	**A-105**
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Roof detailing (Guntur: heat, cyclone wind, short heavy rain)

- Every tile along eaves, verges, hips and ridges clipped or screwed; every third row elsewhere. Cyclone winds lift loose tiles first at the edges.
- Trusses bolted to the RCC ring beam with holding-down bolts; purlins cleated, not just welded tacks.
- Eaves 2'-0" beyond the wall line (kept inside the 27 ft plot on the north) with a drip edge and gutters, so water falls clear of the wall and windows.
- Gutters at both long eaves, downpipes at 8 points into the sump filter, overflow to the road drain; none discharging onto bare soil by the walls.
- Ventilator: hipped tiled cap, 45 deg louvres with mesh, sealed inner shutters, 9 in upstand with flashing (detail on A-503).
- Valleys at the front gable: 600 mm wide GI / aluminium valley gutter under the tiles.
- Attic vented at the ridge and soffits with insect mesh; 50 mm insulation on the deck.
- Leave a 25 mm conduit from roof to DB now for future solar PV on the south slope.

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Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

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Drawing	**Roof plan**
Sheet	**A-106**
Scale	1:75 at A3
Revision	B (pre-construction draft)
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Units	Feet-inches; levels from finished ground

NORTH ELEVATION



SOUTH ELEVATION



Brown line at the eaves = fascia with gutter. Every window has a sunshade (chajja) with a drip groove; sills slope out 1:10.

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Kerala-style single-storey residence

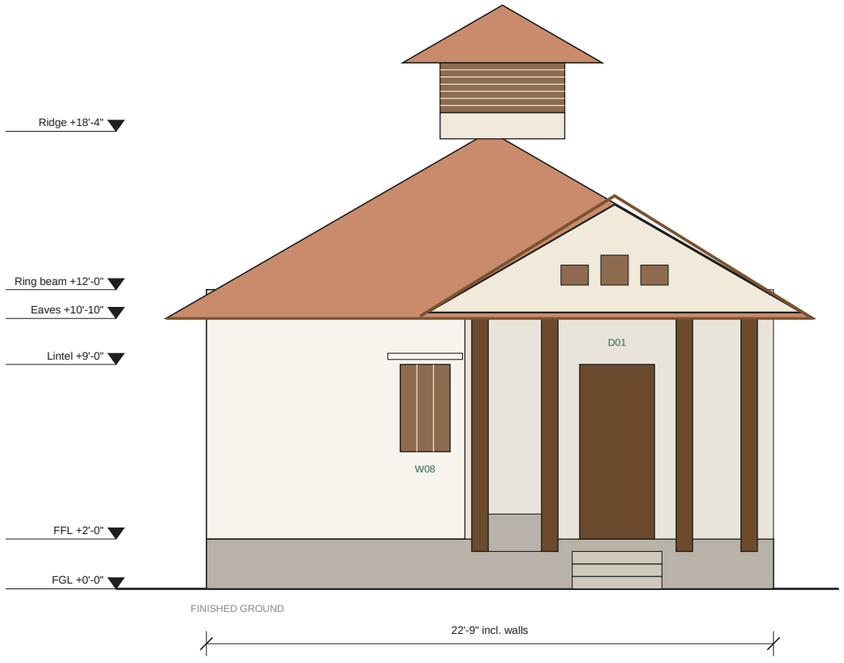
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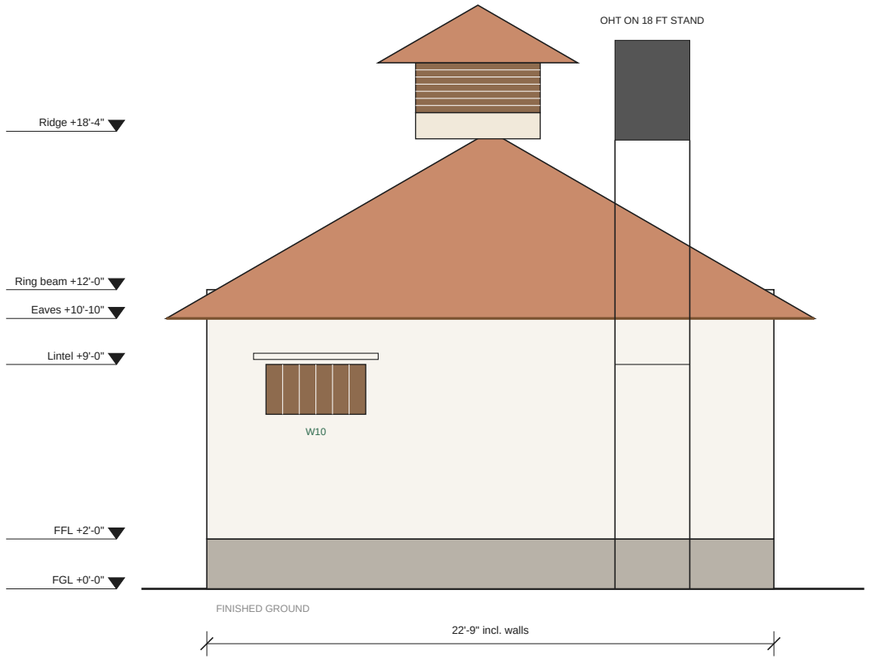
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Stage	Draft for professional review
Drawing	**North and south elevations**
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EAST ELEVATION (ROAD SIDE)



WEST ELEVATION



- Front gable (mukhappu): three timber jali vents let morning light in and hot air out; insect mesh behind.
- Plinth face clad 2 ft in rough stone: the splash zone stays clean and does not erode.
- West wall: no large windows (afternoon sun). Shade with the creeper pergola, not trees, on black cotton soil.
- Entrance: 3 risers of 6 in, 12 in treads, flamed granite with a nosing; foot-wash tap beside the steps.

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Kerala-style single-storey residence

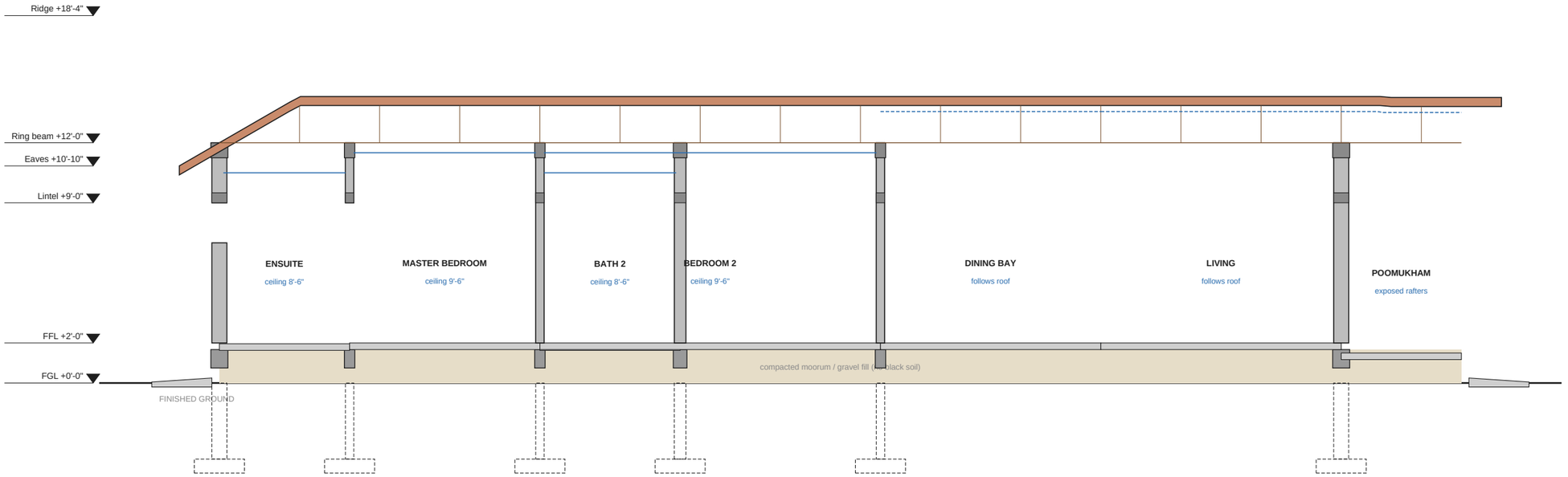
Guntur, Andhra Pradesh

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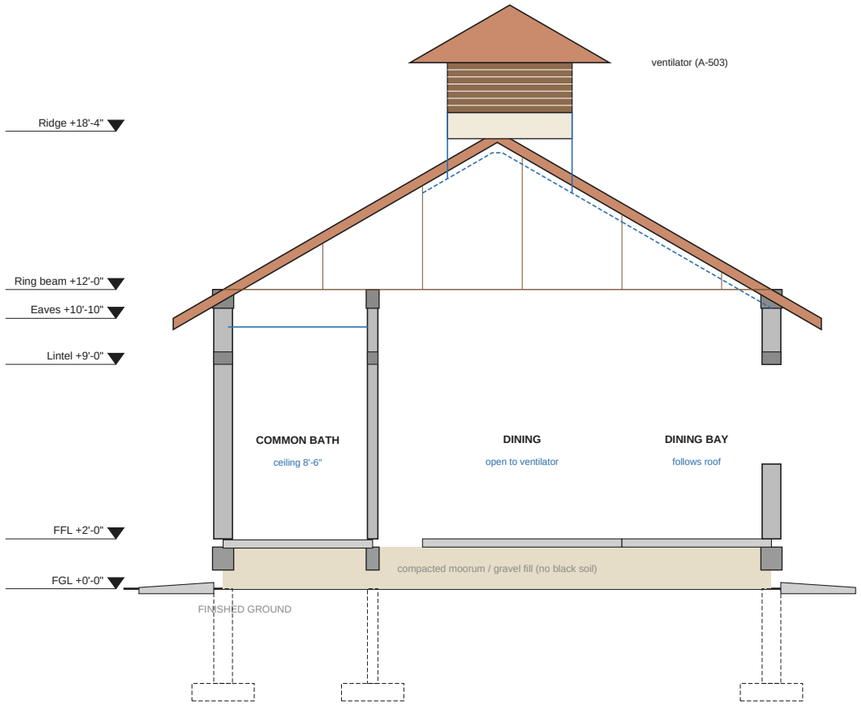
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Stage	Draft for professional review
Drawing	**East (road) and west elevations**
Sheet	**A-202**
Scale	1:75 at A3
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Units	Feet-inches; levels from finished ground

SECTION B-B: LONG SECTION THROUGH BEDROOMS, LIVING AND POOMUKHAM



SECTION A-A: CROSS SECTION THROUGH THE COMMON BATH, DINING AND DINING BAY



Section notes

- Clay tiles on battens, membrane, 50 mm insulation, cement-fibre deck on purlins (attic ventilated at ridge and soffit).
- Steel truss bolted to the RCC ring beam with holding-down bolts (cyclone uplift).
- RCC lintel band at door-head level runs continuously (seismic zone III).
- AAC walls in an RCC frame; mesh over every block-to-concrete joint before plaster.
- Flat false ceilings (blue) in bedrooms, offices, baths and kitchen, with insulation above. Living and dining follow the roof.
- Plinth beam with DPC; plinth fill of compacted moorum, never black soil; anti-termite treatment at every layer.
- Footings dashed and indicative only: type and depth by the engineer after the soil test.
- 3 ft PCC apron sloped away from the wall; joint against the plinth sealed.

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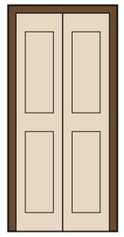
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Stage	Draft for professional review
Drawing	**Sections A-A and B-B**
Sheet	**A-301**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

DOORS (viewed from the side the door swings into; vents above inner doors, 7'-6" to 9'-0"; none at bedrooms or bathrooms)

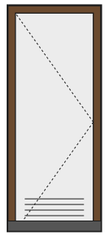
D01



3'-0" x 7'-0"

Teak, double leaf, panelled; 4 in raised threshold outside slope; tower bolts + mortise lock

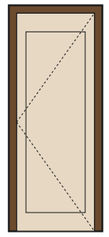
D05



2'-6" x 7'-0"

WPC/FRP shutter, frame on granite base; louvre at bottom
No vent above

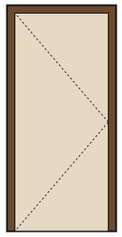
D06



2'-6" x 7'-0"

Teak panelled, single leaf
No vent above

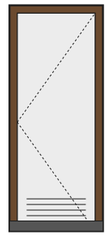
D10



3'-0" x 7'-0"

Flush, teak veneer
No vent above

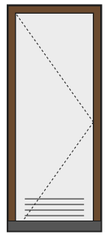
D15



2'-6" x 7'-0"

Flush, waterproof (WPC/FRP), door closer
No vent above

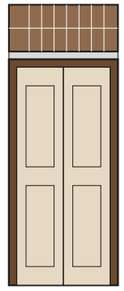
D16



2'-6" x 7'-0"

Flush, waterproof (WPC/FRP)
No vent above

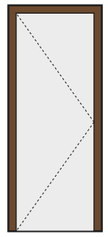
D11



3'-0" x 7'-0"

Carved teak, double leaf (opens inward, clear of the living; faces west, family faces east)
Above: open jali

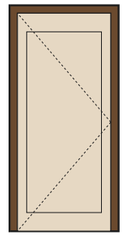
D12



2'-6" x 7'-0"

WPC/FRP shutter, frame on granite base
No vent above

D17

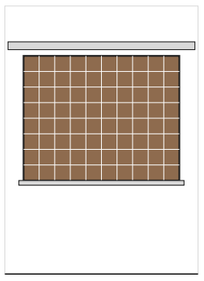


3'-0" x 7'-0"

Teak frame, glazed upper panel, 3 ft (closes off cooking smoke)
No vent above

WINDOWS (viewed from inside; sill and head heights above floor)

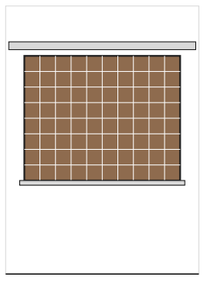
W01



5'-0" wide, sill 3'-0", head 7'-0"

Timber jali + glazed shutters

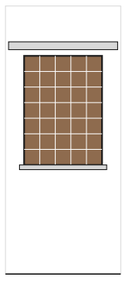
W03



5'-0" wide, sill 3'-0", head 7'-0"

Timber jali + glazed shutters

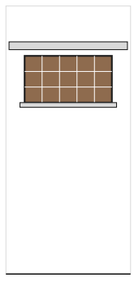
W04



2'-6" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters (over the serving counter)

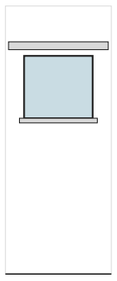
W17



2'-9.5" wide, sill 5'-6", head 7'-0"

High vent to the poomukham, jali + mesh (lamp smoke out)

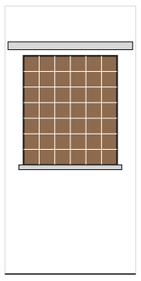
W19



2'-2.5" wide, sill 5'-0", head 7'-0"

Frosted casement, top-hung vent

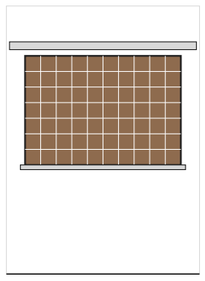
W05



3'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters

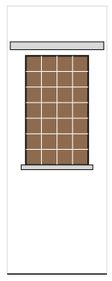
W07



5'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters

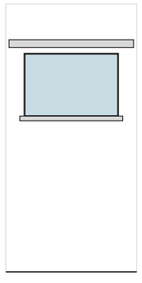
W08



2'-0" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters (2 ft, north end; hob wall kept solid)

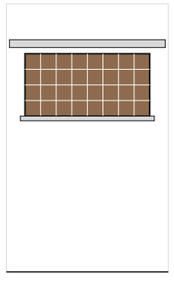
W09



3'-0" wide, sill 5'-0", head 7'-0"

Frosted casement

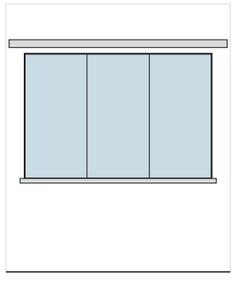
W10



4'-0" wide, sill 5'-0", head 7'-0"

High jali vent

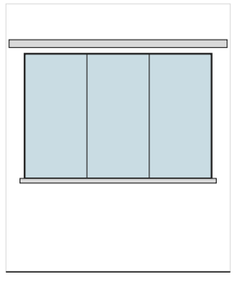
W11



6'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed

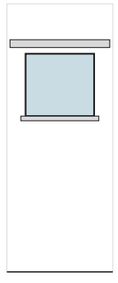
W12



6'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed (6 ft)

W13



2'-2.5" wide, sill 5'-0", head 7'-0"

Frosted casement, top-hung vent

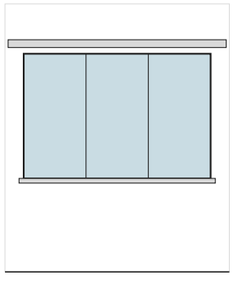
W20



4'-0" wide, sill 7'-6", head 9'-0"

High jali vent to the sitting area, above the bed head (cross-ventilation with W12 and W03)

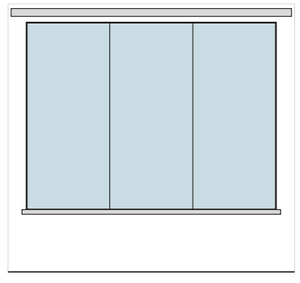
W18



6'-0" wide, sill 3'-0", head 7'-0"

Teak casement, glazed (north light for the dining)

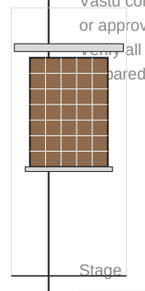
W14



8'-0" wide, sill 2'-0", head 8'-0"

Teak casement, tall, glazed

W16



2'-6" wide, sill 3'-6", head 7'-0"

Timber jali + glazed shutters (over the serving counter)

- All windows: insect mesh (SS or fibreglass) on a separate openable or sliding frame, so glass can be cleaned.
- MS safety grill inside the shutters on every ground-floor window; one escape window per bedroom with an openable grill section.
- Teak is costly and must be seasoned; good alternatives in Guntur: sal or hard pinkado for frames, UPVC for wet-side windows.
- Door frames 5 x 2.5 in; frames in bathrooms and wet areas stand on granite or WPC blocks, never touching the wet floor.
- Hardware: SS 304 hinges and screws (hard water and humidity rust MS); magnetic door stoppers on every door.
- Sunshade (chajja) over every window projects 1'-6" to 2'-0" with a drip groove under the front edge.

Kerala Home, Phase 1

Kerala-style single-storey residence

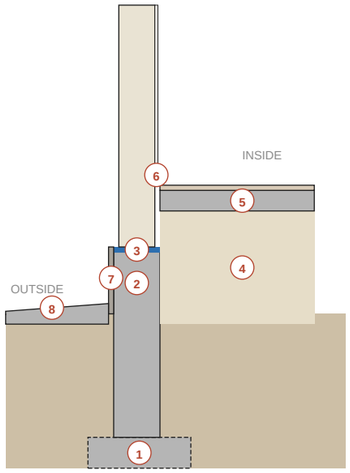
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

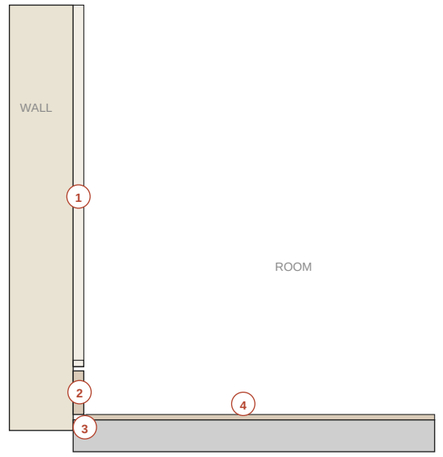
Stage	Draft for professional review
Drawing	**Door and window details**
Sheet	**A-401**
Scale	1:60 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

1 Plinth, DPC, floor and apron



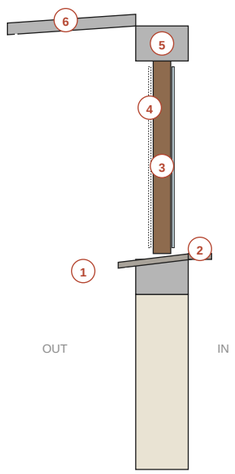
1 Footing by engineer after soil test; on black cotton soil often 1.5-2.5 m deep or under-reamed piles.
2 RCC plinth beam ties all columns; 75 mm sand cushion under it on expansive soil.
3 DPC 40 mm with waterproof compound at plinth top, under the first AAC course.
4 Moorum / gravel fill compacted in 150 mm layers; anti-termite treatment at each layer.
5 PCC bed 100 mm with polythene sheet, then screed and tile.
6 Skirting 4 in, flush with plaster face (plaster stops in a groove): no dust ledge.
7 Stone cladding on the plinth face to 2 ft: rain splash zone stays clean.
8 3 ft PCC apron sloping 1:30 away; polysulphide joint at the wall so water never reaches the foundation.

2 Flush skirting at floor-wall joint



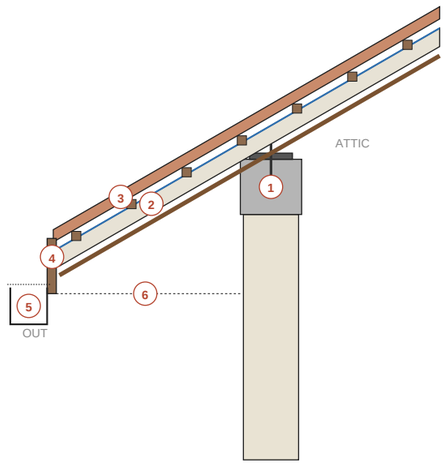
1 Plaster stops 1/4 in above skirting in a clean groove.
2 Skirting tile 4 in high set in the plaster line, face flush with the wall.
3 5 mm gap filled with silicone / PU sealant: allows movement, no cracked tiles.
4 In wet rooms use a cove tile or wall tile down to the floor with epoxy grout instead.

3 Window sill, frame and sunshade



1 External sill slopes out 1:10 and projects 2 in past the plaster with a drip groove underneath.
2 Internal sill in stone, 1/2 in higher than the outer sill (water cannot run inward).
3 Frame on a bed of sealant; gap to wall filled with PU foam and sealed both sides.
4 Insect mesh on a separate openable frame.
5 RCC lintel band continuous around the house at door-head level.
6 Sunshade 1'-6" to 2'-0", top sloped out, drip groove under the front edge (no black streaks on walls).

4 Eaves, gutter and roof build-up



1 RCC ring beam with holding-down bolts cast in; truss shoe bolted, not just welded.
2 Cement-fibre deck + 50 mm insulation + breathable membrane under the battens.
3 Mangalore tiles; eave, verge, hip and ridge tiles clipped or screwed (cyclone wind).
4 Fascia board with drip edge flashing into the gutter.
5 150 mm gutter with leaf guard, falls 1:300 to downpipes.
6 Ventilated soffit with insect / bird mesh: hot attic air escapes at the ridge.

Kerala Home, Phase 1

Kerala-style single-storey residence

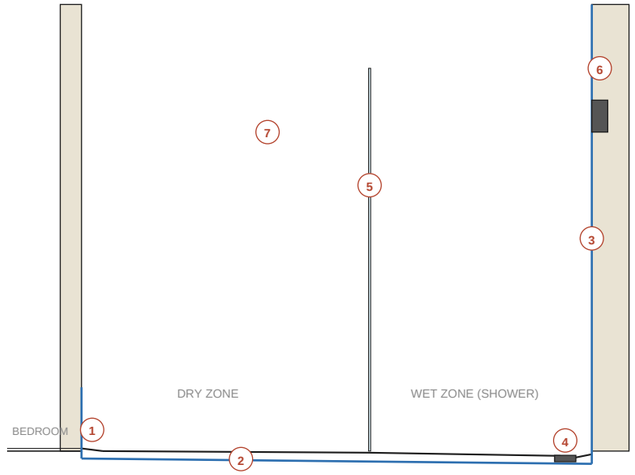
Guntur, Andhra Pradesh

Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

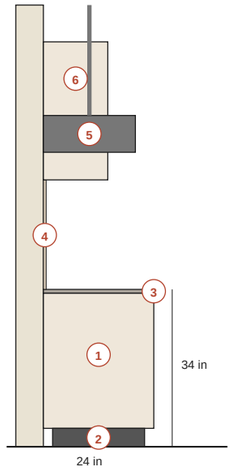
Stage	Draft for professional review
Drawing	**Typical details 1: plinth, floor, window, eaves**
Sheet	**A-501**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

5 Bathroom: wet and dry zones



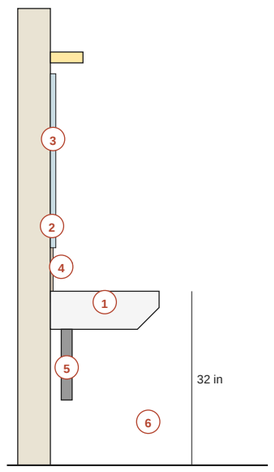
1 Bathroom 12 mm below the bedroom floor with a sloped granite threshold: no raised step to trip on.
2 Waterproof membrane across the floor and 12 in up all walls; 48 h flood test before tiling.
3 Shower walls waterproofed to 7 ft; tiles to ceiling in the shower, 7 ft elsewhere.
4 Linear drain at the far wall; only the wet zone falls (1:60). Dry zone stays dry and safe.
5 8 mm toughened glass partition, 6 in clear of the floor drain line; matt fittings hide hard-water marks.
6 Concealed diverter and pipes in the 7 in wet wall; isolation valve outside the bathroom.
7 Exhaust fan high on the outer wall, opposite the door, ducted outside (not into the attic).

6 Kitchen counter and chimney



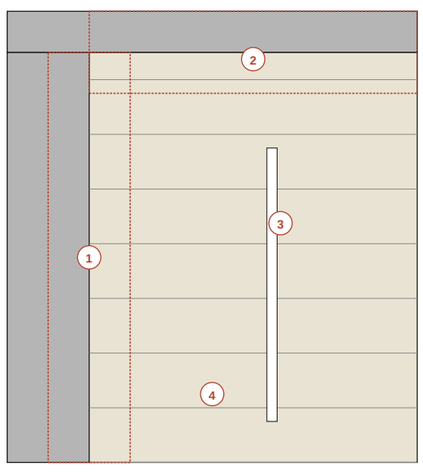
1 Counter top at 34 in (865 mm) for standing cooking; 30-31 in section for a chapati / grinding area.
2 Base units on 4 in legs or a granite plinth: floor can be mopped, no termites, no water damage.
3 20 mm granite (local Chimakurthy / Ongole) with a drip edge; sink under the window, hob on the east run.
4 Stone or tile between counter and wall cabinets, full height; no painted wall behind the hob.
5 Chimney 26-30 in above the hob, ducted straight out through the wall (no long bends).
6 Wall cabinets 14 in deep, starting 22-24 in above the counter; plenty of 16 A sockets at 42 in.

7 Hand-wash basin (serving counter)



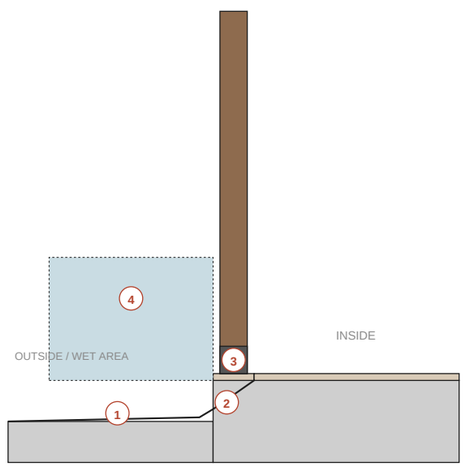
1 Wall-hung basin, rim at 32 in (810 mm); bottle trap, so the floor below can be cleaned.
2 Tiled or stone splash-back 2 ft above the basin: painted walls stain and peel here.
3 Mirror from 40 in to 72 in, light above (not behind) the mirror, socket at 42 in kept 2 ft from the tap.
4 Soap / shelf niche recessed into the wall, sloped forward so water does not pool.
5 Waste pipe into the wall, not across the floor; a floor trap is not needed for a dining-area basin.
6 Children: keep a fold-down step; tap with a long spout so water lands in the bowl, not on the counter.

8 AAC wall joints and chases



1 Galvanised mesh 150 mm wide, centred on every AAC-to-column and AAC-to-beam joint, before plaster.
2 Top course: 20 mm gap under the beam packed with PU foam / flexible filler, not rigid mortar.
3 Chases cut with a wall chaser, max 1/3 of block depth; refill with polymer mortar + mesh.
4 Thin-bed polymer mortar 3 mm, blocks laid dry; first course on the DPC in normal mortar to level.

9 Door thresholds and rain kerbs



1 Outside floor 3-6 in lower than inside and sloping away 1:50: rain never flows in.
2 Granite threshold with a 1 in upstand / weather bar under external doors.
3 Wet-area door frames stand on a granite or WPC block; wood never touches a wet floor.
4 Open wet area: floor 3 in lower and sloping out to the walkway, so rain drains away from the kitchen door.

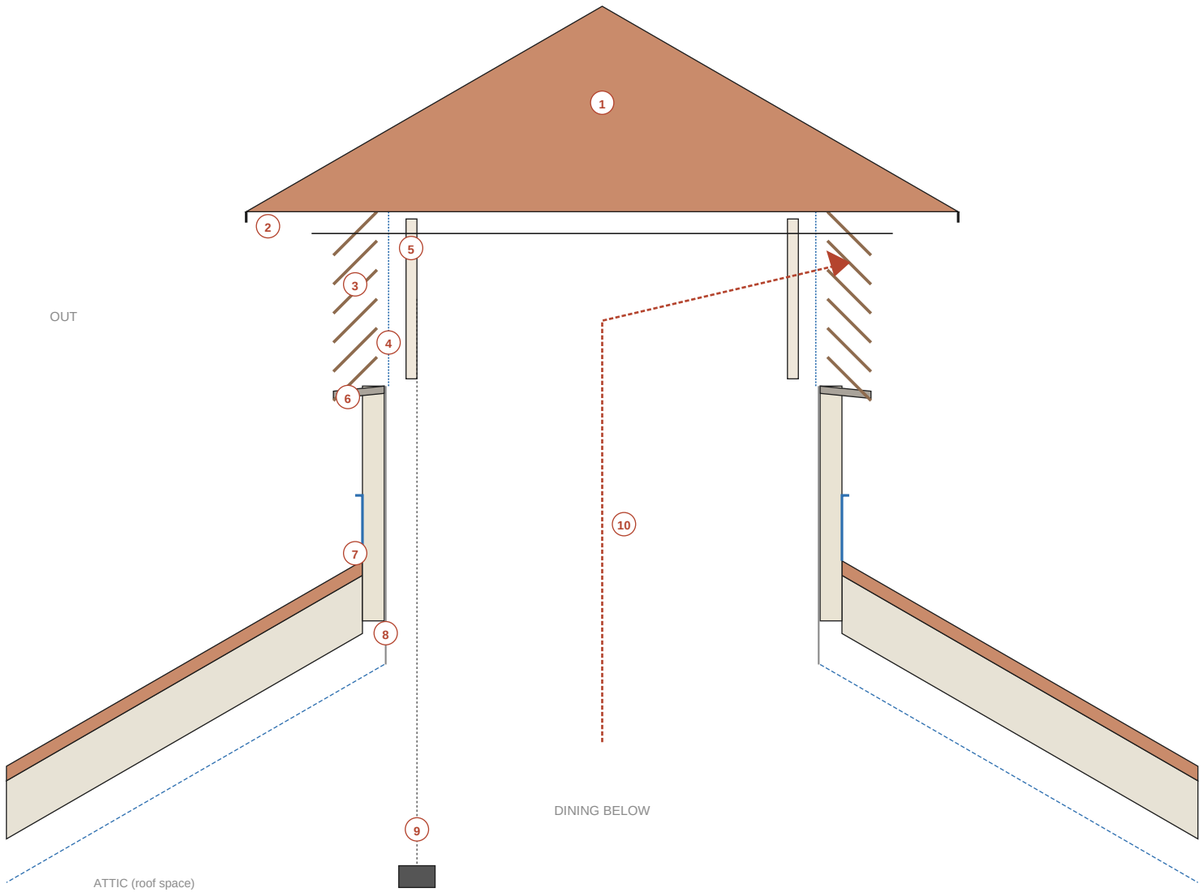
Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

Pre-construction draft for review by the architect, structural engineer, MEP and Vastu consultant. Not for construction or approval until checked and signed. Verify all dimensions and levels on site. Prepared with AI assistance (Claude).

Stage	Draft for professional review
Drawing	**Typical details 2: bath, kitchen, basin, AAC**
Sheet	**A-502**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

10 Dining ventilator: section



1 Hipped cap at the main roof pitch (30 deg), clay tiles clipped, insulated board under: sheds rain, nothing ponds.
2 Cap overhangs the louvres by 1'-6" with a drip edge, so rain falls well clear of them.
3 Louvre blades at 45 deg sloping down and out with a 4 in lap: no straight line through for wind-driven rain.
4 SS 304 insect mesh about 1.2 mm (18 x 16) in a continuous frame, sealed at every corner: this is what stops mosquitoes.
5 Inner shutters with EPDM rubber seals: closed for AC, cool nights and dust storms. Opening them gives access to clean the mesh.
6 Sill sloped out 1:10 with a drip; any water that gets past the louvres runs back out.
7 9 in minimum upstand above the tiles with lead / membrane flashing and counter-flashing.
8 Shaft lined with insulated fibre-cement board down to the dining ceiling: attic heat and dust stay out.
9 Winding opener (or cord and pulley) with the handle 5 ft above the dining floor; no ladder needed.
10 Hot air from the dining and living rises up the shaft and leaves through the louvres; fresh air enters low through south jalis and windows.

How it works

- Hot air collects under the sloped ceilings of the dining and living. The ventilator is the highest opening in the house, so that air rises out through it.
- Replacement air comes in low and cool through the south jalis, the open wet area and the windows.
- It works best on still, hot evenings, when there is no breeze to cross-ventilate the house.
- No glass, so no direct sun or greenhouse heat on the dining table. Daylight in the dining comes from the living, the nook window and the hall.

Using it

- Summer days and evenings: shutters open.
- AC running in the living: shutters closed, or the cool air leaves through the roof.
- Monsoon storms and cyclone warnings: shutters closed.
- Dust storms or winter nights: shutters closed.
- Twice a year: open the shutters and brush or vacuum the mesh.

Before it is closed up

- Engineer to frame the ventilator into the trusses and check cyclone uplift on the cap.
- Water-test the louvres and flashing with a hose before the dining ceiling is closed.
- Check the mesh frame for gaps with a torch from inside before the shutters are fitted.
- Size: 5 x 5 ft louvre box, 2 ft louvre band, about 20 sq ft of free vent area.

Kerala Home, Phase 1

Kerala-style single-storey residence
Guntur, Andhra Pradesh
Plot 27' x 75', road on east

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Stage	Draft for professional review
Drawing	**Typical details 3: dining ventilator**
Sheet	**A-503**
Scale	NTS
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

Room schedule

ID	Room	Centreline	Clear	Area sq ft (clear)
R01	Workstation (open, walls optional later)	9'-0" x 8'-0"	8'-7.5" x 7'-7.5"	66
R02	Master bay (old store, open to the bedroom)	7'-0" x 8'-0"	6'-9.5" x 7'-7.5"	52
R03	Dressing alcove	6'-6" x 6'-0"	6'-1.5" x 5'-8.5"	35
R04	Ensuite	6'-6" x 8'-0"	5'-11" x 7'-4"	43
R05	Master bedroom	9'-6" x 14'-0"	9'-1" x 13'-7.5"	124
R08	Bath 2 (attached to Bedroom 2)	7'-0" x 7'-0"	6'-6" x 6'-4"	41
R07	Common bath	6'-6" x 6'-0"	6'-0" x 5'-5"	32
R09	Bedroom 2	17'-0" x 14'-6"	16'-7" x 13'-11"	231
R10	Dining	11'-0" x 8'-0"	10'-9.5" x 8'-0"	86
R19	Dining bay (north window)	11'-0" x 6'-0"	10'-9.5" x 5'-7.5"	61
R13	Wet area (open to the walkway)	5'-0" x 6'-0"	4'-6" x 5'-8.5"	26
R14	Sitting and serving area (open)	20'-6" x 7'-6"	20'-1" x 6'-11"	139
R15	Kitchen	14'-0" x 10'-0"	13'-5" x 9'-3"	124
R16	Living	12'-0" x 12'-0"	11'-7.5" x 11'-3"	131
R11	Pooja	4'-0" x 4'-0"	3'-5" x 3'-5"	12
R17	Poomukham (verandah)	6'-0" x 12'-0"	5'-7.5" x 11'-7.5"	65
	Total (clear)			1267

Door schedule (heights 7'-0"; vents 7'-6" to 9'-0")

ID	Location	Width	Type	Vent above
D01	Main entrance (centred under the front gable)	3'-0"	Teak panelled, double leaf	-
D05	Ensuite (opens outward)	2'-6"	Flush, waterproof (WPC/FRP)	-
D06	Master suite entry, from the sitting area into the master bay	2'-6"	Teak panelled, single leaf	-
D10	Bedroom 2 (from the sitting area, east end)	3'-0"	Flush, teak veneer	-
D15	Common bath (from the sitting area, opens outward, hinged at the north jamb so the open leaf stays clear of the counter)	2'-6"	Flush, waterproof (WPC/FRP), door closer	-
D16	Bath 2 (east wall, from Bedroom 2, opens outward)	2'-6"	Flush, waterproof (WPC/FRP)	-
D11	Pooja	3'-0"	Carved teak, double leaf (opens inward, clear of the living; faces west, family faces east)	Open jali
D12	Kitchen to wet area	2'-6"	Steel or WPC, lockable (the wet area is open to outside)	-
D17	Kitchen entry (west, from the dining alcove; hinged at the north jamb)	3'-0"	Teak frame, glazed upper panel, 3 ft (closes off cooking smoke)	-

Window and vent schedule (sill/head above floor)

ID	Room	Wall	Width	Sill	Head	Type
W01	Workstation	S	5'-0"	3'-0"	7'-0"	Timber jali + glazed shutters
W03	Sitting and serving area	S	5'-0"	3'-0"	7'-0"	Timber jali + glazed shutters
W04	Basin and crockery nook	S	2'-6"	3'-6"	7'-0"	Timber jali + glazed shutters (over the serving counter)
W17	Pooja	E	2'-9.5"	5'-6"	7'-0"	High vent to the poomukham, jali + mesh (lamp smoke out)
W19	Common bath	S	2'-2.5"	5'-0"	7'-0"	Frosted casement, top-hung vent
W05	Kitchen	S	3'-0"	3'-6"	7'-0"	Timber jali + glazed shutters
W07	Kitchen	S	5'-0"	3'-6"	7'-0"	Timber jali + glazed shutters
W08	Kitchen	E	2'-0"	3'-6"	7'-0"	Timber jali + glazed shutters (2 ft, north end; hob wall kept solid)
W09	Ensuite	N	3'-0"	5'-0"	7'-0"	Frosted casement
W10	Ensuite	W	4'-0"	5'-0"	7'-0"	High jali vent
W11	Master bedroom	N	6'-0"	3'-0"	7'-0"	Teak casement, glazed
W12	Bedroom 2	N	6'-0"	3'-0"	7'-0"	Teak casement, glazed (6 ft)
W13	Bath 2	N	2'-2.5"	5'-0"	7'-0"	Frosted casement, top-hung vent
W20	Bedroom 2	-	4'-0"	7'-6"	9'-0"	High jali vent to the sitting area, above the bed head (cross-ventilation with W12 and W03)
W18	Dining bay	N	6'-0"	3'-0"	7'-0"	Teak casement, glazed (north light for the dining)
W14	Living	N	8'-0"	2'-0"	8'-0"	Teak casement, tall, glazed
W16	Basin and crockery nook	S	2'-6"	3'-6"	7'-0"	Timber jali + glazed shutters (over the serving counter)
SK1	Roof	-	-	-	-	Ventilator over dining, 5 x 5 ft, no glass: hipped tiled cap at 30 deg with 1'-6" overhang; 2 ft band of 45 deg louvres on all four sides; SS insect mesh behind; sealed inner shutters on a winding opener; 9 in upstand; lined shaft to the dining ceiling (see A-503)
GV1 -3	Roof	-	-	-	-	Three glazed/jali vents in front gable (mukhappu), morning light into living

Finishes

Space	Floor	Walls / skirting	Ceiling
Living, dining, hall	Red oxide (IPS) or polished Kota / Tandur stone	4 in flush skirting in floor stone; lime-finish or mineral paint	Sloped, follows roof, insulated board under deck
Bedrooms, offices, dressing	Matt vitrified 2x2 ft or red oxide	4 in flush skirting; putty + washable emulsion	False ceiling at 9'-6", 50 mm insulation above
Pooja	White marble or teak	Teak panelling behind idol; 4 in marble skirting	Teak / board at 9'-6"
Kitchen	Anti-skid matt vitrified or Kota, mid-tone	Tile / stone full height between counter and wall cabinets	Moisture-resistant board at 9'-6"
Ensuite, Bath 2, common bath	Anti-skid ceramic (R11), falls 1:60 to drain	Tiles to 7 ft; full height in shower; waterproofing 12 in up all walls, 7 ft in shower	Moisture-resistant board / PVC at 8'-6"
Wet area	Rough Cuddapah / Kota stone, falls to drain	4 ft stone or tile dado	Open truss
Family store, sitting area	Red oxide or vitrified	4 in flush skirting; washable paint	Board at 9'-6"
Poomukham, steps	Flamed / leather-finish granite (not polished)	Granite skirting; plinth face clad 2 ft (splash zone)	Exposed rafters + board
Exterior	Plinth apron: PCC 3 ft wide sloped away	AAC + 20 mm plaster with waterproof additive; elastomeric paint	-
Roof (all)	Mangalore clay tiles, every edge tile clipped / screwed	Membrane + 50 mm insulation on cement-fibre deck	Steel truss on RCC ring beam

Kerala Home, Phase 1

Kerala-style single-storey residence

Guntur, Andhra Pradesh

Plot 27' x 75', road on east

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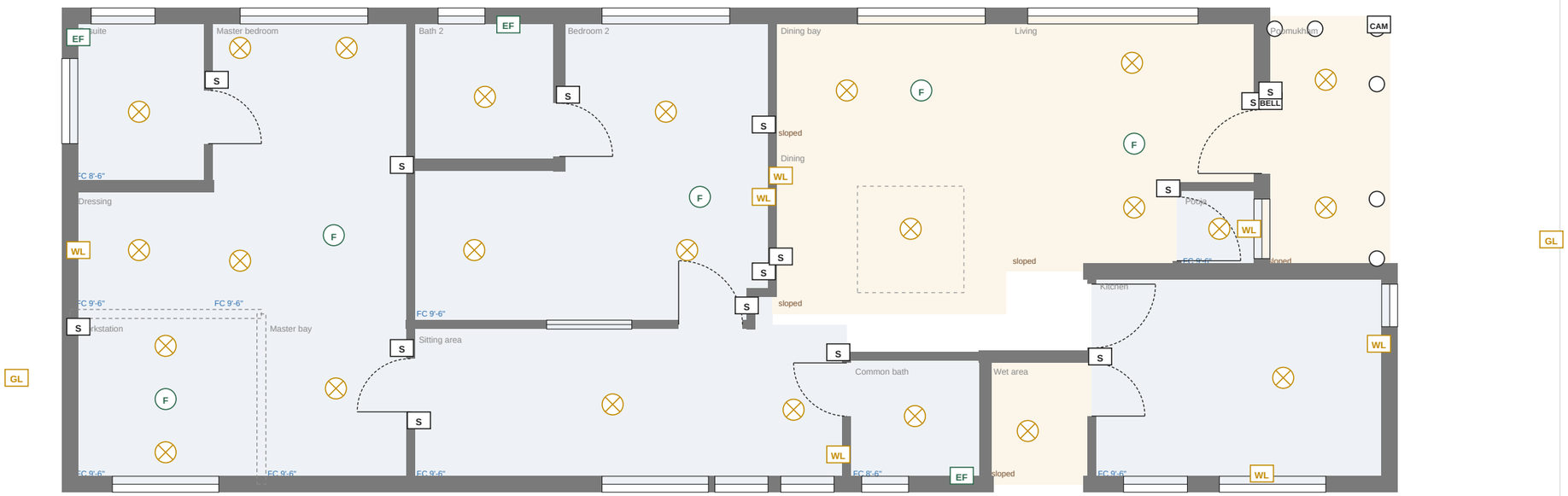
Stage	Draft for professional review
Drawing	**Schedules**
Sheet	**A-601**
Scale	-
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

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Lighting schedule

Code	Item	Height	No.
L	Ceiling light	ceiling	24
WL	Wall light / mirror light	6'-6"	7
F	Ceiling fan point	ceiling	5
EF	Exhaust fan	7'-0"	3
S	Switchboard	4'-0"	15
BELL	Bell push	4'-6"	1
CAM	CCTV point	8'-0"	1
GL	Outdoor / gate light	7'-0"	2

Lighting and ceiling notes

- Switchboards on the latch side of every door at 4 ft; 2-way switches at both bedroom beds and both ends of the hall.
- Bathroom switches outside the door; geyser switch with indicator lamp outside too.
- Fans at least 7 ft clear of the floor; long downrods in sloped ceilings; fan hooks cast or bolted, never in false ceiling board.
- Warm 2700-3000 K light in living and bedrooms, 4000 K in kitchen and offices; no downlights directly over the bed head.
- False ceilings (blue) with 50 mm insulation above; access panel near every light cluster and near AC pipe runs.
- Lights, fans, office sockets and router on the inverter circuit; outdoor lights on a timer or photocell.

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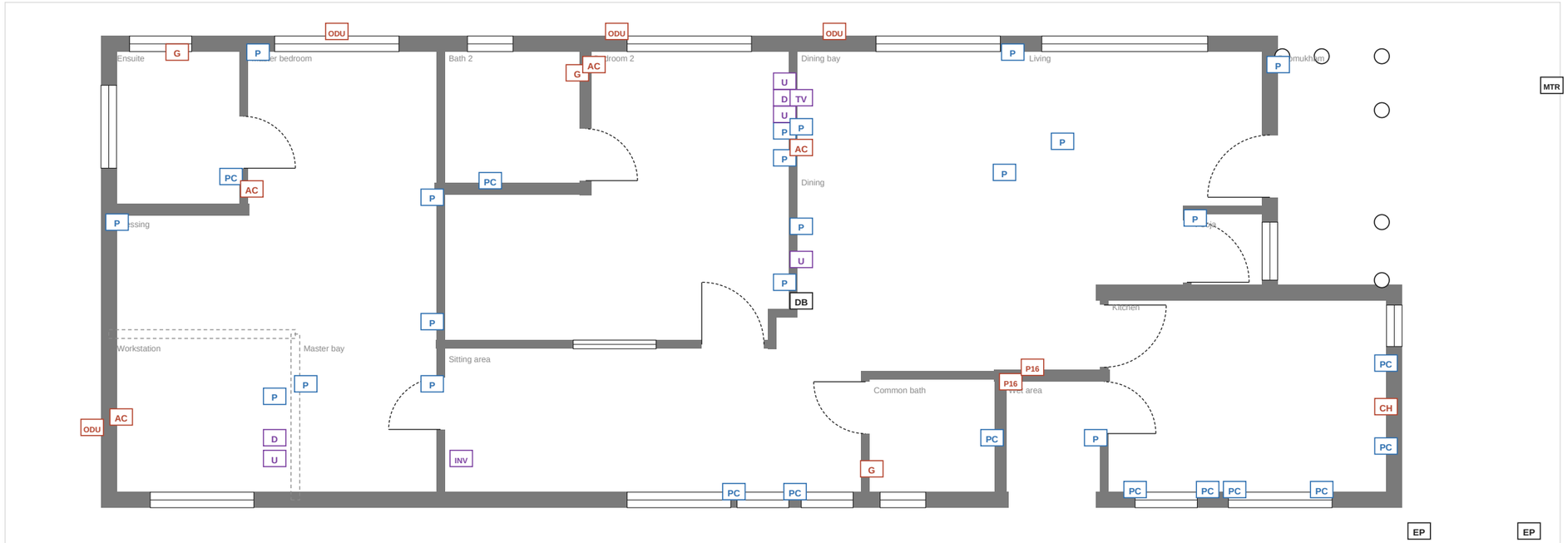
Stage	Draft for professional review
Drawing	**Lighting, fans and ceilings**
Sheet	**E-101**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground

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Power schedule

Code	Item	Height	No.
CH	Kitchen chimney point (16 A)	7'-0"	1
P	6 A socket	1'-0"	18
PC	6 A socket at counter	3'-6"	11
P16	16 A power socket	varies	2
U	Inverter (UPS) socket	1'-0"	4
D	Data point (Cat6)	1'-0"	2
TV	TV + data point	3'-6"	1
AC	AC indoor unit (16 A socket, sleeve, drain)	7'-6"	4
ODU	AC outdoor unit bracket	7'-0" outside	4
G	Geyser (16 A)	6'-6"	3
DB	Distribution board	5'-0" to top	1
INV	Inverter + battery	floor	1
MTR	Meter at gate	5'-0"	1
EP	Earth pit	ground	2

Power notes

- Separate MCB circuit for each AC, geyser, kitchen power, washing machine and pump; RCCB 30 mA on all socket circuits.
- AC: 16 A socket at 7'-6", 65 mm sleeve sloping out, drain pipe to a gully (not onto the wall); ODUs on shaded north / west walls on brackets.
- AC indoor units never above the bed head or desk chair: cold draught on the head is the common complaint.
- Work desks (Office 2, Bedroom 2): 2 x UPS sockets + data + 1 normal socket at desk height; Cat6 from the router to the TV and both desks.
- Kitchen: 5-6 counter sockets at 42 in, 16 A for the fridge, chimney and microwave / OTG; none within 2 ft of the sink.
- DB on the dining face of Bedroom 2's east wall (central, dry) with 25% spare ways; meter at the gate; 2 earth pits; surge protector.
- Empty 25 mm conduits from the roof (solar PV), gate (CCTV, bell, intercom) and sump (pump controller) to the DB.

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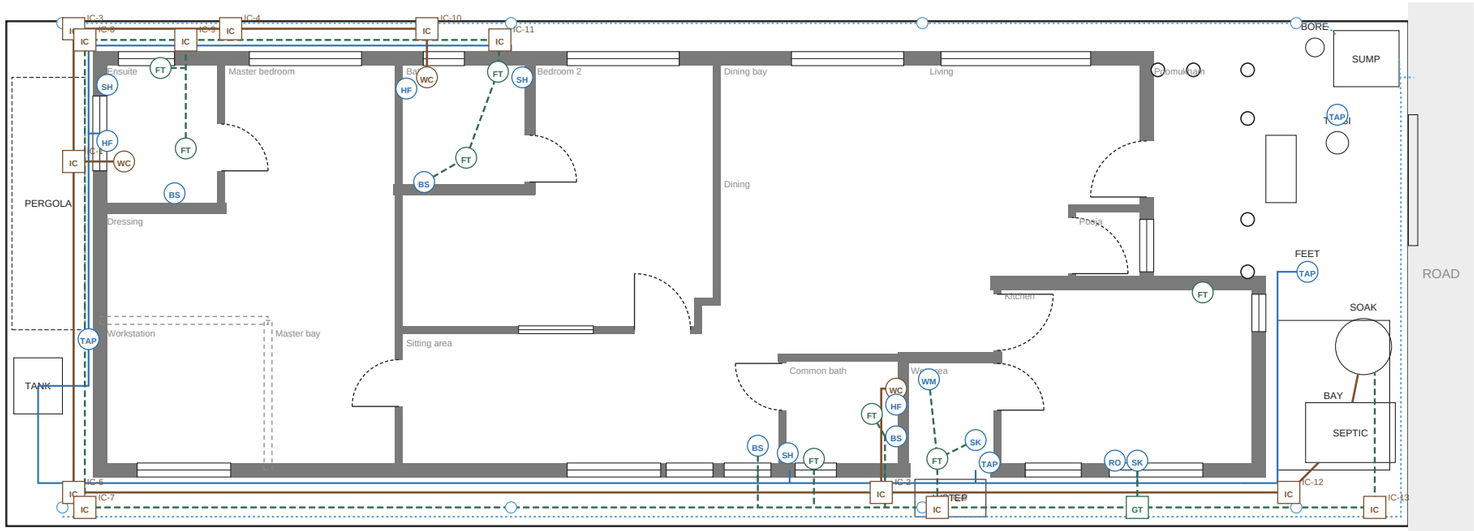
Stage	Draft for professional review
Drawing	**Power, AC and data**
Sheet	**E-102**
Scale	1:75 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
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Water supply (CPVC), from overhead tank

Soil line (uPVC 110 mm) to septic tank

Grey water (uPVC 75 mm) to the soak pit

Rainwater to sump filter, overflow to road

Fixtures and points

Code	Item	No.
WC	Wall-hung WC, concealed cistern	3
HF	Health faucet	3
BS	Wash basin	4
SH	Shower mixer (hot + cold)	3
FT	Floor trap / linear drain	8
SK	Sink	2
WM	Washing-machine inlet + drain	1
RO	RO purifier point	1
TAP	Tap	4
IC	Inspection chamber	13
GT	Grease trap	1

Plumbing notes

- Overhead tank feeds by gravity; isolation valve for each bathroom and the kitchen, outside the room.
- Supply pipes run on the outside of walls in clamps or shallow chases, never under the floor slab.
- Soil and grey water kept separate; inspection chamber at every bend and junction; 1:40 falls on 110 mm, 1:30 on 75 mm.
- Grease trap before the kitchen line joins anything; vent pipe from each WC stack above the eaves.
- Septic tank (sealed) and a lined soak pit in the front yard, as far from the house and sump as the plot allows; wet black cotton soil swells and cracks the house, so prefer treated outflow to a road drain if one is available.
- Rain from both gutters to the sump through a first-flush filter; overflow to the road drain.
- Test the bore / supply water: RO for drinking, softener for bathrooms if hardness is high; matt or PVD fittings.

Stage	Draft for professional review
Drawing	**Water supply and drainage**
Sheet	**P-101**
Scale	1:100 at A3
Revision	B (pre-construction draft)
Date	30 Oct 2026
Units	Feet-inches; levels from finished ground